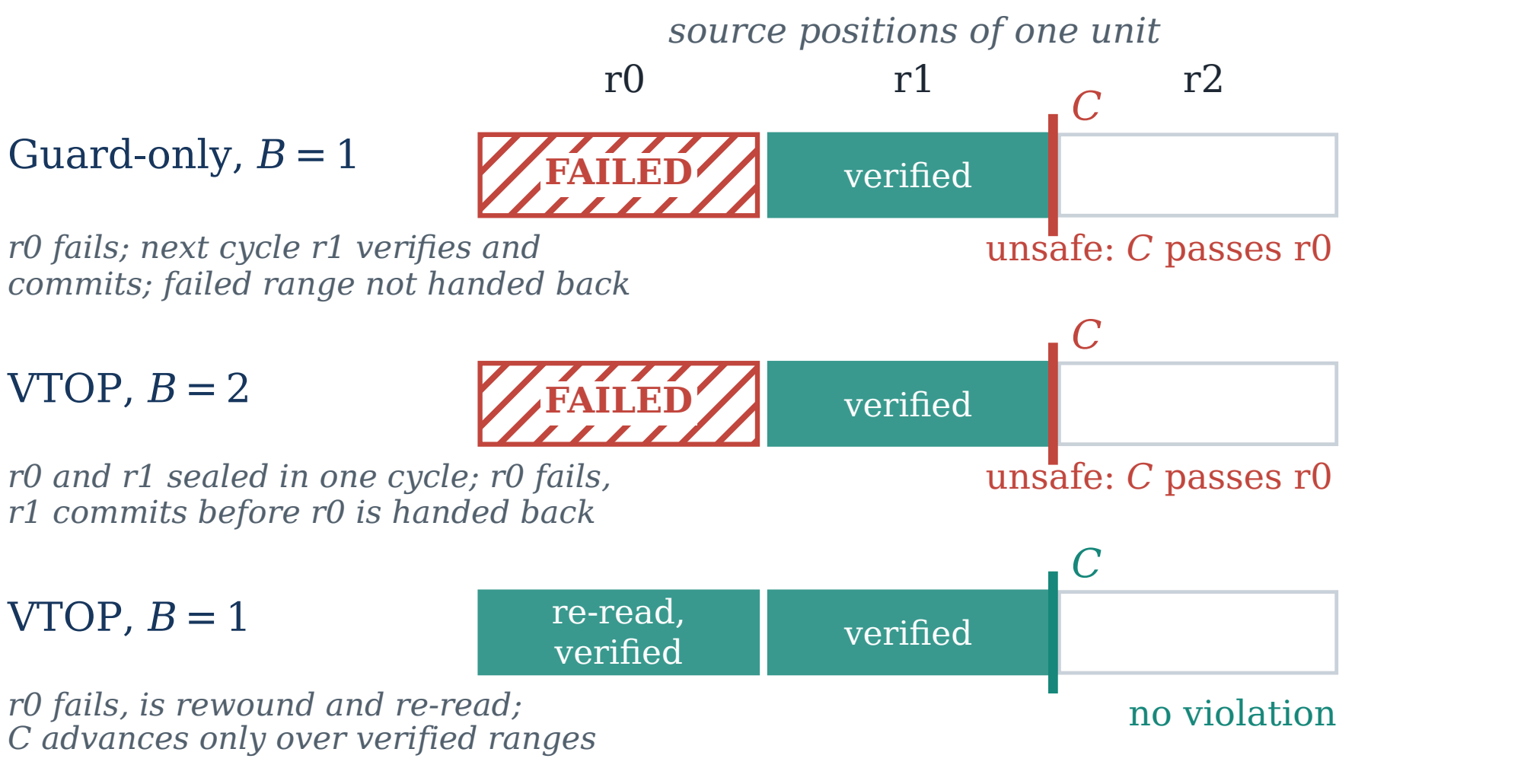
source positions of one unit
r0
r1
r2
Guard-only, $B = 1$
FAILED
verified
C
unsafe: C passes r0
r0 fails; next cycle r1 verifies and commits; failed range not handed back
VTOP, $B = 2$
FAILED
verified
C
unsafe: C passes r0
r0 and r1 sealed in one cycle; r0 fails, r1 commits before r0 is handed back
VTOP, $B = 1$
re-read, verified
verified
C
no violation
r0 fails, is rewound and re-read; C advances only over verified ranges